

can influence aggregate inflation measures, making it important to account for these interactions. By leveraging information across hierarchical levels, these models produce robust forecasts that address both short-term volatility and long-term trends. Building on this foundation, we chose to explore another hierarchical model that enables bidirectional information propagation, further enhancing prediction accuracy and capturing more intricate relationships within the data.

Conventional hierarchical forecasting methods generally use top-down, bottom-up, or middle-out strategies. In the top-down approach, forecasts are generated at the highest level of the hierarchy and then allocated to lower levels. In contrast, the bottom-up approach aggregates forecasts from the lowest level to produce predictions for higher levels. The middle-out approach combines elements of both top-down and bottom-up methods [Hyndman and Athanasopoulos, 2011].

Recent advancements have incorporated machine learning techniques into hierarchical forecasting. Models like hierarchical RNNs (HRNNs) and hierarchical LSTMs exploit the hierarchical structure of data to capture dependencies both within and across different levels. These models have demonstrated improved performance across various applications, such as sales forecasting and energy demand prediction [Wickramasuriya et al., 2019]. By leveraging information from multiple levels of the hierarchy, they offer more accurate and reliable predictions compared to traditional non-hierarchical approaches.

2.3 Hierarchical Long Short-Term Memory Network

Hierarchical Long Short-Term Memory networks (Hierarchical LSTMs) are an extension of traditional LSTM models, designed to handle data with multiple levels of structure, such as hierarchical time series or sequential data organized into nested groups. These models capture dependencies both within and across different levels of the hierarchy by processing information at each level of granularity. For example, in time series forecasting, a hierarchical LSTM can model trends at higher levels (e.g., overall sales) while simultaneously capturing detailed patterns at lower levels (e.g., sales by region or product category). This architecture allows the model to account for complex interactions across levels, improving prediction accuracy and robustness in tasks where hierarchical relationships are important [Lin et al., 2022].

2.4 Hierarchical Attention Network

Hierarchical Attention Networks (HANs) are a deep learning framework designed to model hierarchical structures within data, particularly useful in natural language processing tasks. Unlike standard attention mechanisms, HANs apply attention at different levels of the data hierarchy, such as words within sentences and sentences within documents. This enables the model to focus on the most relevant components at each level, improving its ability to capture contextual relationships and interactions across the hierarchy. For example, in

document classification, HANs can first highlight key words within individual sentences, then identify the most important sentences across the document. This hierarchical attention approach not only boosts interpretability but also enhances performance in tasks that rely on understanding multi-level data structures. The effectiveness of HANs has made them essential in various NLP applications, including document classification, sentiment analysis, and text summarization [Yang et al., 2016].

3 BiHRNN: Bi-Directional Hierarchical Recurrent Neural Network

Hierarchical data refers to structured data organized into multiple levels, with each level representing a different degree of aggregation or detail. In the context of inflation, hierarchical data can range from the overall inflation index, such as the Consumer Price Index (CPI), down to disaggregated components like regional indices or individual product categories. This multi-level structure allows for analysis at varying levels of granularity, capturing relationships and dependencies across different layers. Such an approach is particularly valuable for inflation modeling, as changes at lower levels, such as specific goods or services, can propagate upward and influence aggregate measures, enabling more accurate and comprehensive forecasting.

3.1 Hierarchical Recurrent Neural Networks

Before delving into the specifics of the BiHRNN model, we provide a brief overview of its predecessor, the HRNN model [Barkan et al., 2023]. The HRNN model is specifically designed to address the challenges of inflation forecasting in hierarchically structured datasets, where lower levels are often characterized by data sparsity and heightened volatility in change rates. To enhance predictions, the HRNN propagates information from parent categories to node categories within the hierarchy by employing hierarchical Gaussian priors. This approach connects each node’s parameters to its parent’s, allowing the model to share information across levels of aggregation. By leveraging parent-level information, the HRNN mitigates the effects of sparse or noisy data at finer levels and ensures consistency in forecasts across the hierarchy. Furthermore, it utilizes RNNs, specifically Gated Recurrent Units (GRUs) [Dey and Salemt, 2017], which incorporate a feedback loop, enabling predictions to account for temporal dependencies. This combination of hierarchical priors and temporal modeling makes the HRNN particularly effective for capturing both cross-level interactions and dynamic patterns in inflation data.